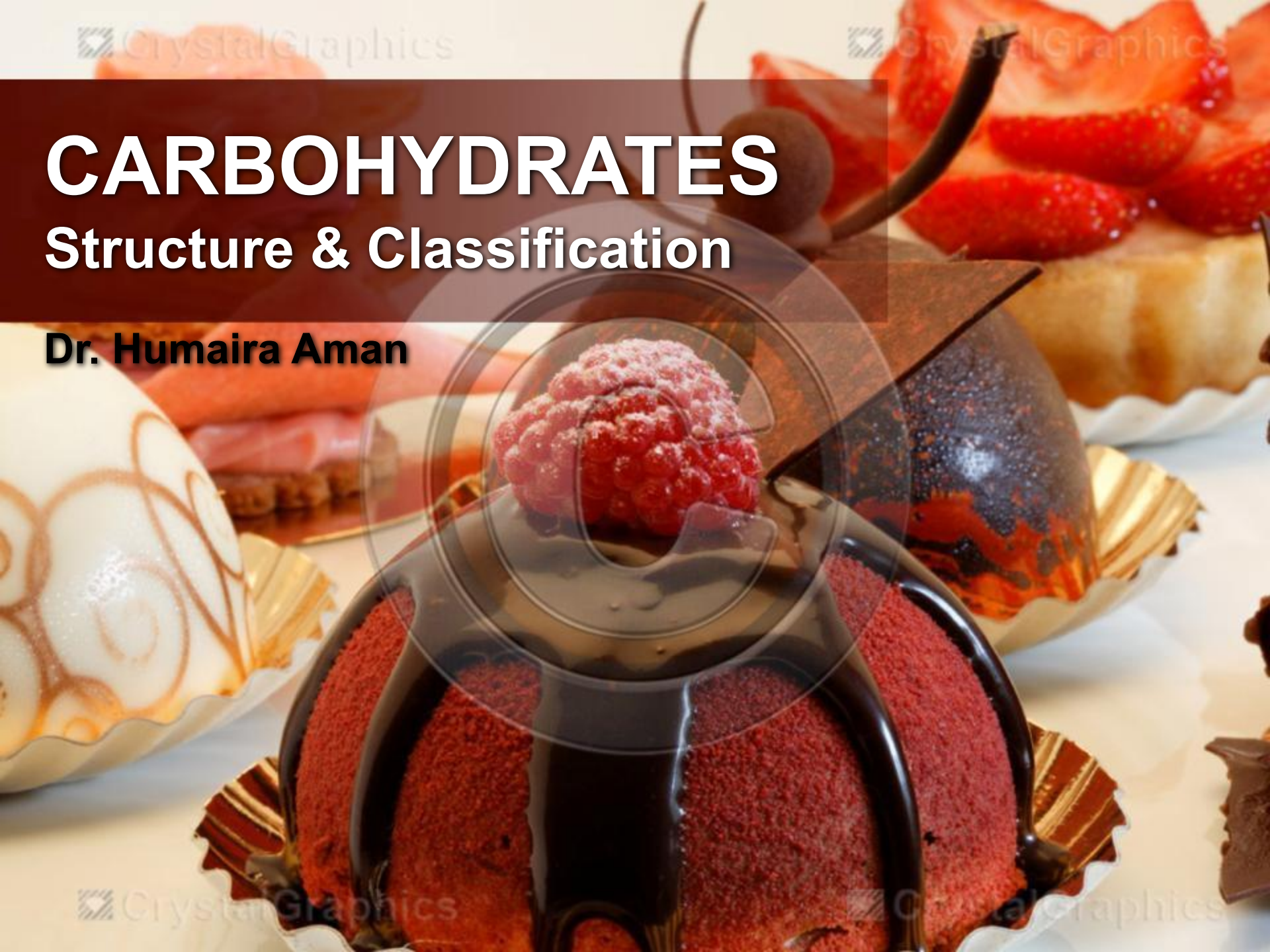


CARBOHYDRATES

Structure & Classification

Dr. Humaira Aman





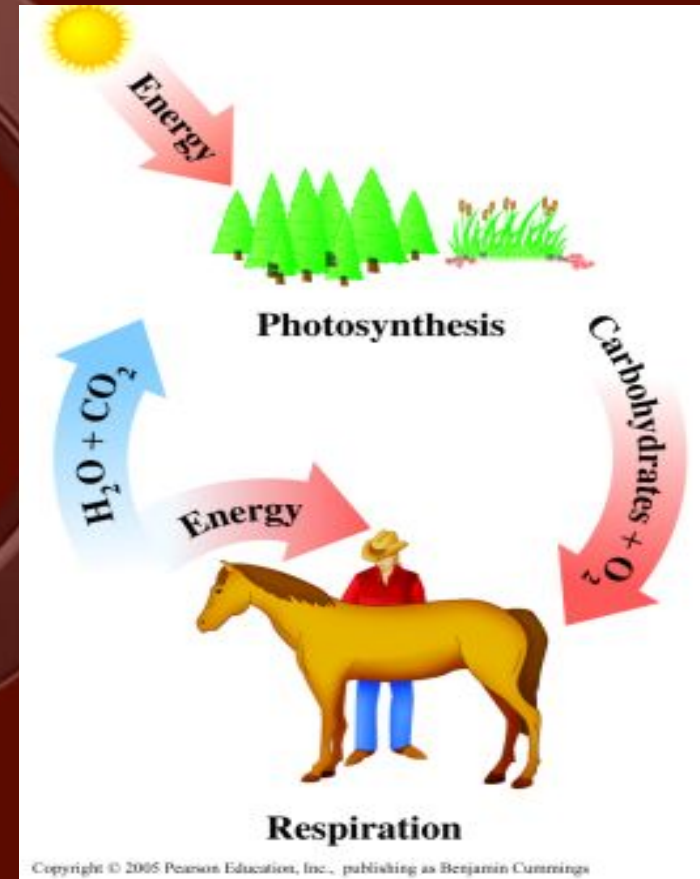
OBJECTIVES

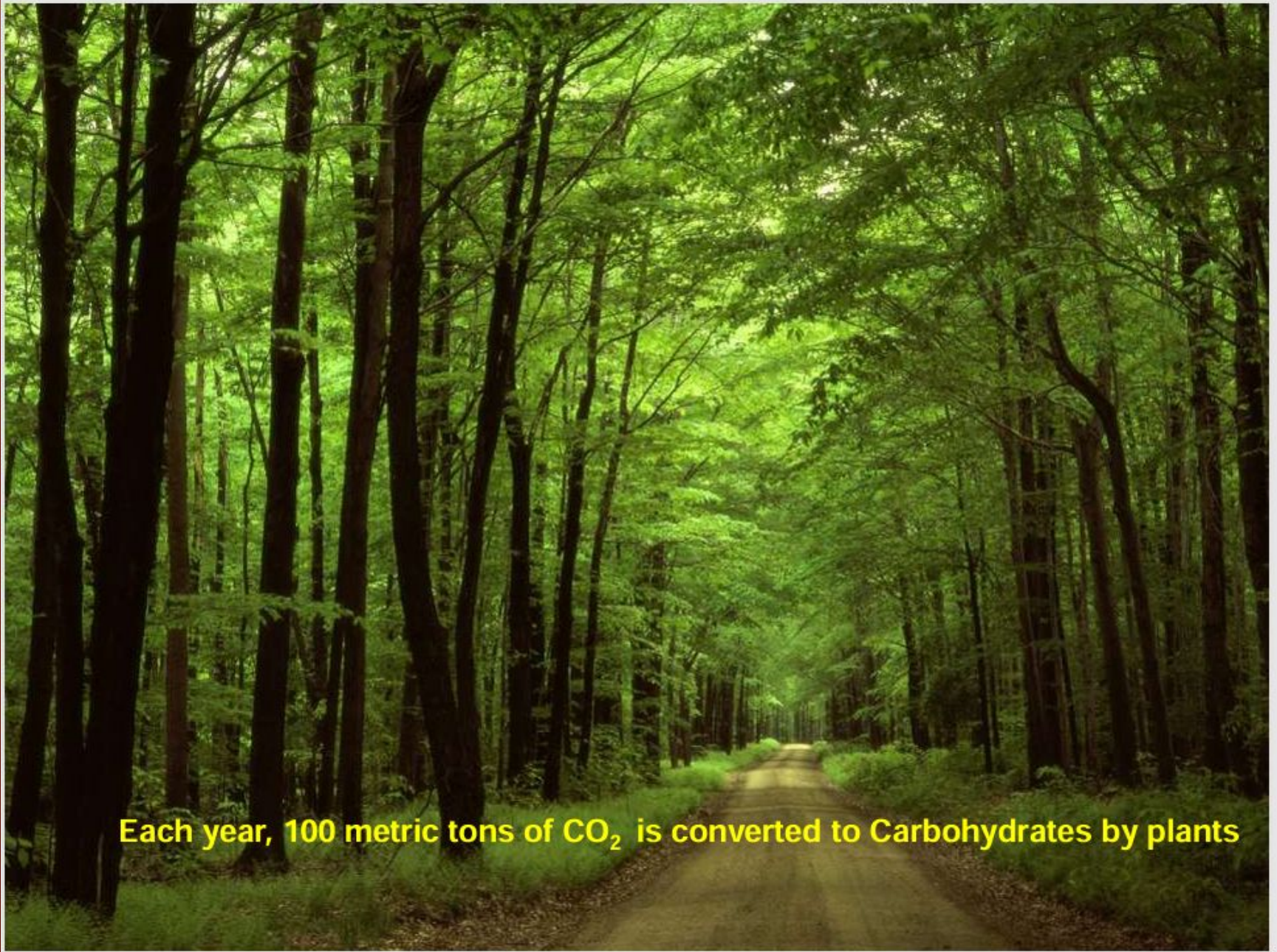
- At the end of the lecture, student would be able to:
- Define carbohydrates
- Classify them
- Know about isomers, epimers and enantiomers.

Carbohydrates

also called saccharides

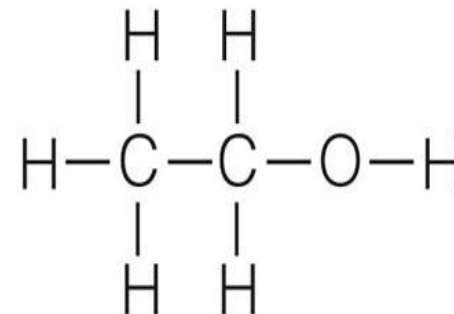
- The term Carbohydrates is applied to a large number of relatively heterogeneous compounds found in all animals and plants.
- They are most abundant biomolecules on earth.





Each year, 100 metric tons of CO_2 is converted to Carbohydrates by plants

Cont..

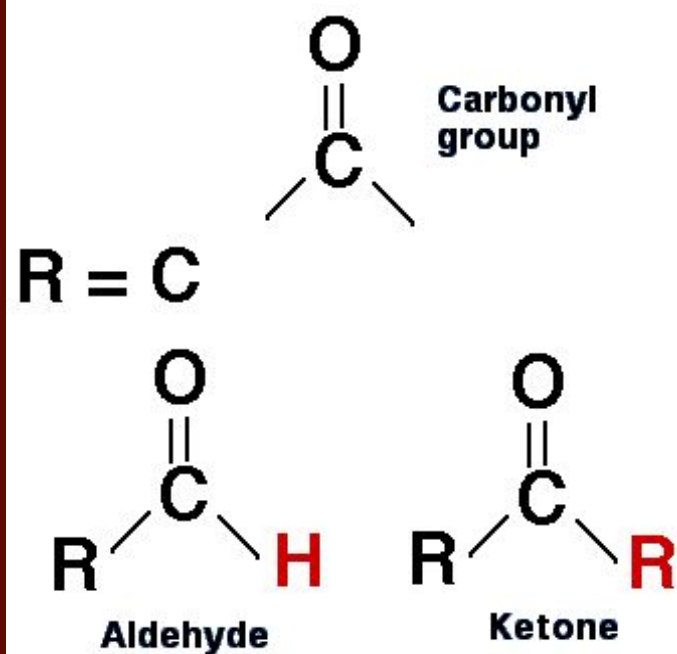
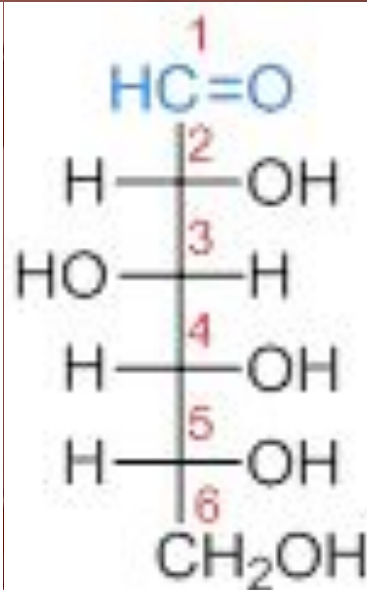


- Hydrates of carbon, composed of carbon, hydrogen and oxygen.
- Def: “They are polyhydroxylated compounds with at least 3 C atoms with potentially active carbonyl groups which may either be aldehyde or ketone groups”.
- They also include those compounds which yield them on hydrolysis.

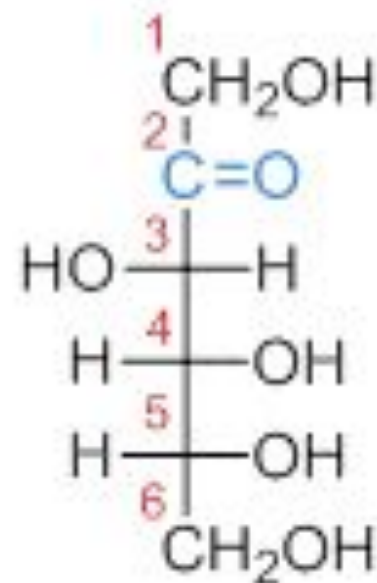
cont..

OR

- Carbohydrates can also be defined as “polyhydroxyaldehydes or ketones or compounds which produce them on hydrolysis”.

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Glucose
(Aldose)



Fructose
(Ketose)

cont..

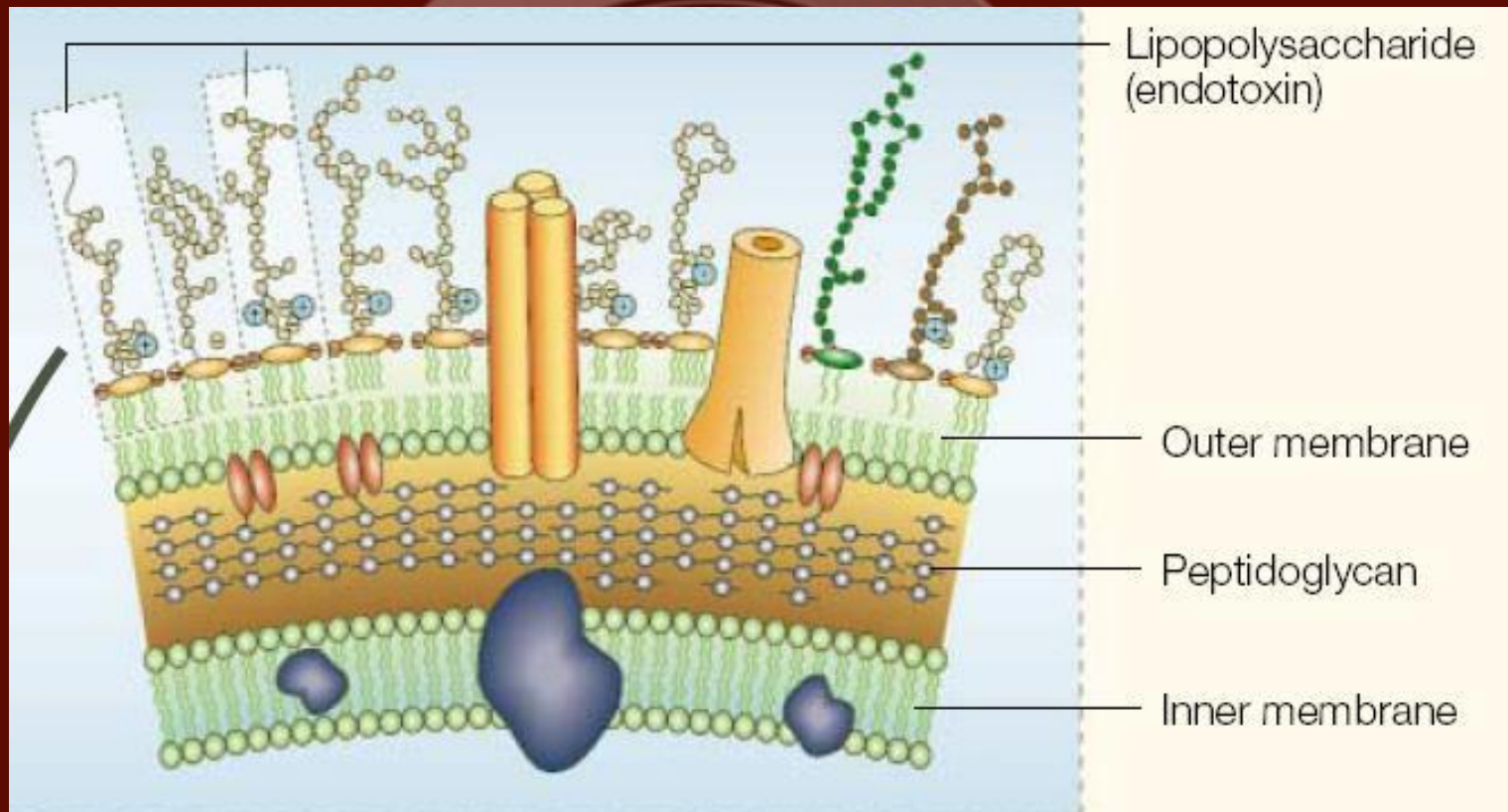
- The general molecular or empirical formula
- $C_n (H_2O)_n$.
- Some carbohydrates contain Sulphur, Nitrogen or Phosphorus
- Exceptions are acetic acid $C_2H_4O_2$ and lactic acid $C_3H_6O_3$.

Biochemical role or Functions of Carbohydrates

- They have a wide range of functions, including:
 - Providing a significant fraction of the dietary calories for most organisms.
 - Main sources of ENERGY in body (4kcal/g)
 - RBCs and Brain cells have an absolute requirement of carbohydrates.
 - Acting as a storage form (starch and glycogen) of energy in the body.



- Serving as cell membrane components.
 - _ Glycoproteins and glycolipids are components of cell membranes and receptors.
- Serve as a structural component of many organisms.
 - e.g. Cellulose in plants, exoskeleton of insects, cell wall of microbes, mucopolysaccharides and ground substance in higher organisms.



Classification of Carbohydrates

Carbohydrates

```
graph TD; C[Carbohydrates] --> M[Monosaccharides<br/>1 sugar unit<br/>e.g. Glucose, fructose etc]; C --> D[Disaccharides<br/>2 sugar units<br/>e.g. Sucrose]; C --> O[Oligosaccharides<br/>3-9 units<br/>e.g. Maltotriose]; C --> P[Polysaccharides<br/>>10]; P --> H[Homoglycans<br/>e.g. starch, glycogen]; P --> R[Heteroglycans<br/>e.g. GAGs or glycosaminoglycans];
```

Monosaccharides
1 sugar unit

e.g. Glucose,
fructose etc

Disaccharides
2 sugar units
e.g. Sucrose

Oligosaccharides
3-9 units
e.g. Maltotriose

Polysaccharides
>10

Homoglycans
e.g. starch,
glycogen

Heteroglycans
e.g. GAGs or
glycosaminoglycans

Classification of Carbohydrates

- **Monosaccharides (simple sugar, one unit)**
- **Disaccharides**
Contain 2 monosaccharide units
- **Oligosaccharides**
Contain from three to about nine monosaccharides
- **Polysaccharides**
Contain many (more than ten) monosaccharide units

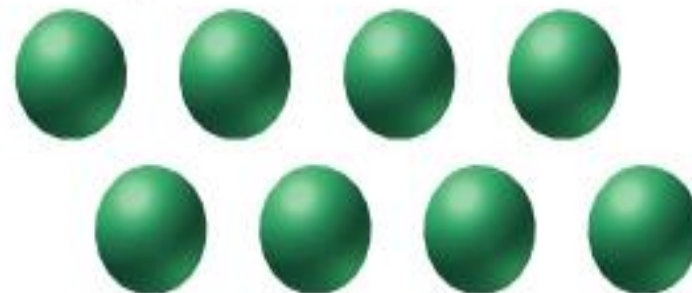
Monosaccharide + H₂O $\xrightarrow{H^+}$ no hydrolysis



Disaccharide + H₂O $\xrightarrow{H^+}$ two monosaccharide units



Polysaccharide + many H₂O $\xrightarrow{H^+}$ many monosaccharide units



Timberlake, *General, Organic, and Biological Chemistry*. Copyright © Pearson Education Inc., publishing as Benjamin Cummings

Monosaccharides

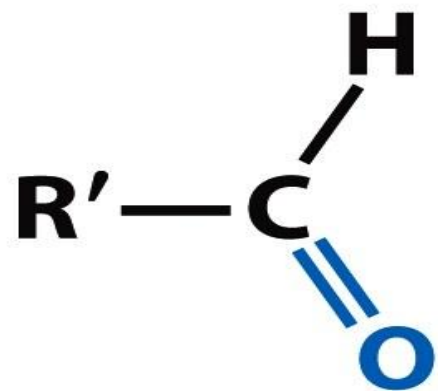
- Three Carbons = Trioses
- Four Carbons = Tetroses
- Five Carbons = Pentoses
- Six Carbons = Hexose

Monosaccharides

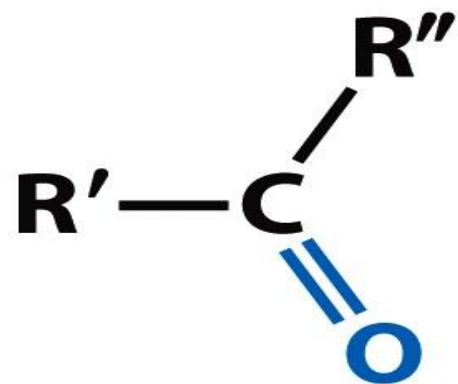
Monosaccharides consist of

- 3-6 carbon atoms typically.
- a carbonyl group (aldehyde or ketone).
- several hydroxyl groups.
- 2 types of monosaccharide structures:
Aldoses and ketoses

Aldoses & Ketoses



Aldehyde



Ketone

Sugars made from
aldehydes form
Aldoses

Sugars made from
ketones form
ketoses

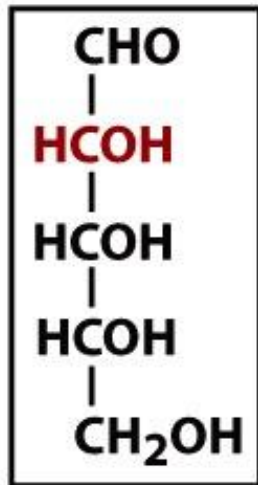
Monosaccharides

- Molecules having only one actual or potential sugar group

No. of carbon	Generic name	Aldoses	Ketoses
3	Trioses	Aldotriose e.g. glyceraldehyde	Ketotriose e.g. Dihydroxyacetone
4	Tetroses	Aldotetrose e.g. Erythrose	Ketotetrose e.g. Erythrulose
5	Pentoses	Aldopentoses e.g. Arabinose, Xylose, Ribose	Ketopentoses e.g. Xylulose, Ribulose
6	Hexoses	Aldohexose e.g. Glucose, Galactose, Mannose	Ketohexose e.g. Fructose
7	Heptoses	Aldoheptose: Glucoheptose	Ketoheptose e.g. Sedoheptulose

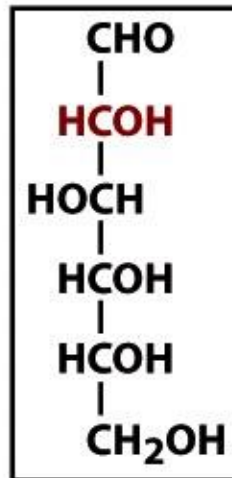
Important Aldoses To Study

6-C sugars

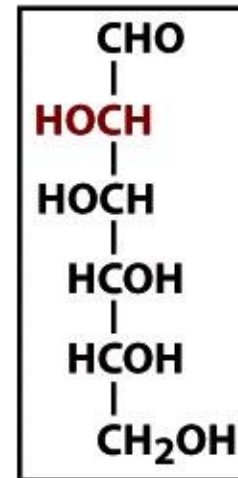


D-Ribose (Rib)

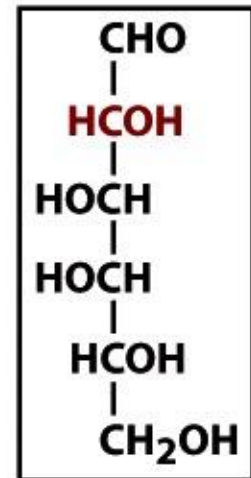
**5-C
sugar**



**D-Glucose
(Glc)**



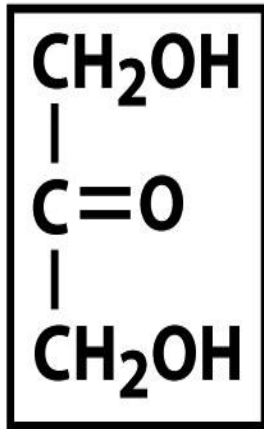
**D-Mannose
(Man)**



**D-Galactose
(Gal)**

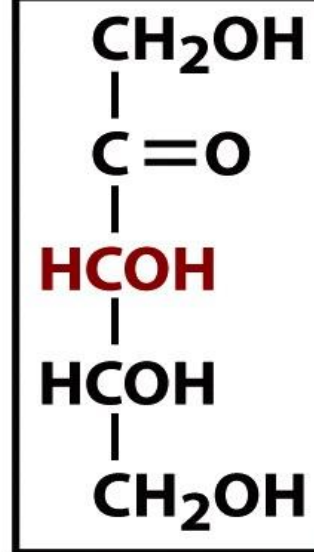
All begin with CHO group

Important Ketoses To Study



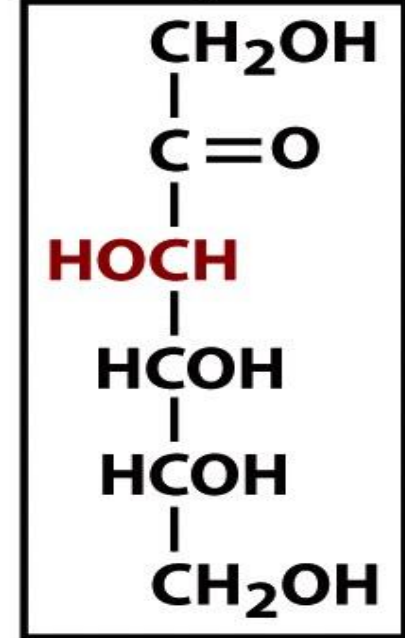
Dihydroxyacetone

**3-C
sugar**



D-Ribulose

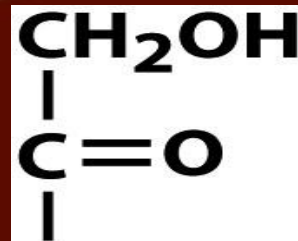
**5-C
sugar**



D-Fructose

**6-C
sugar**

Ketoses begin with



group

Pentoses of Physiological importance

Sugar	Where Found	Biochemical Importance	Clinical Significance
D-Ribose	Nucleic acids.	Structural elements of nucleic acids and coenzymes, eg, ATP, NAD, NADP, flavo-proteins. Ribose phosphates are intermediates in pentose phosphate pathway.	
D-Ribulose	Formed in metabolic processes.	Ribulose phosphate is an intermediate in pentose phosphate pathway.	
D-Arabinose	Gum arabic. Plum and cherry gums.	Constituent of glycoproteins.	
D-Xylose	Wood gums, proteoglycans, glycosaminoglycans.	Constituent of glycoproteins.	
D-Lyxose	Heart muscle.	A constituent of a lyxoflavin isolated from human heart muscle.	

Hexoses of Physiological Importance

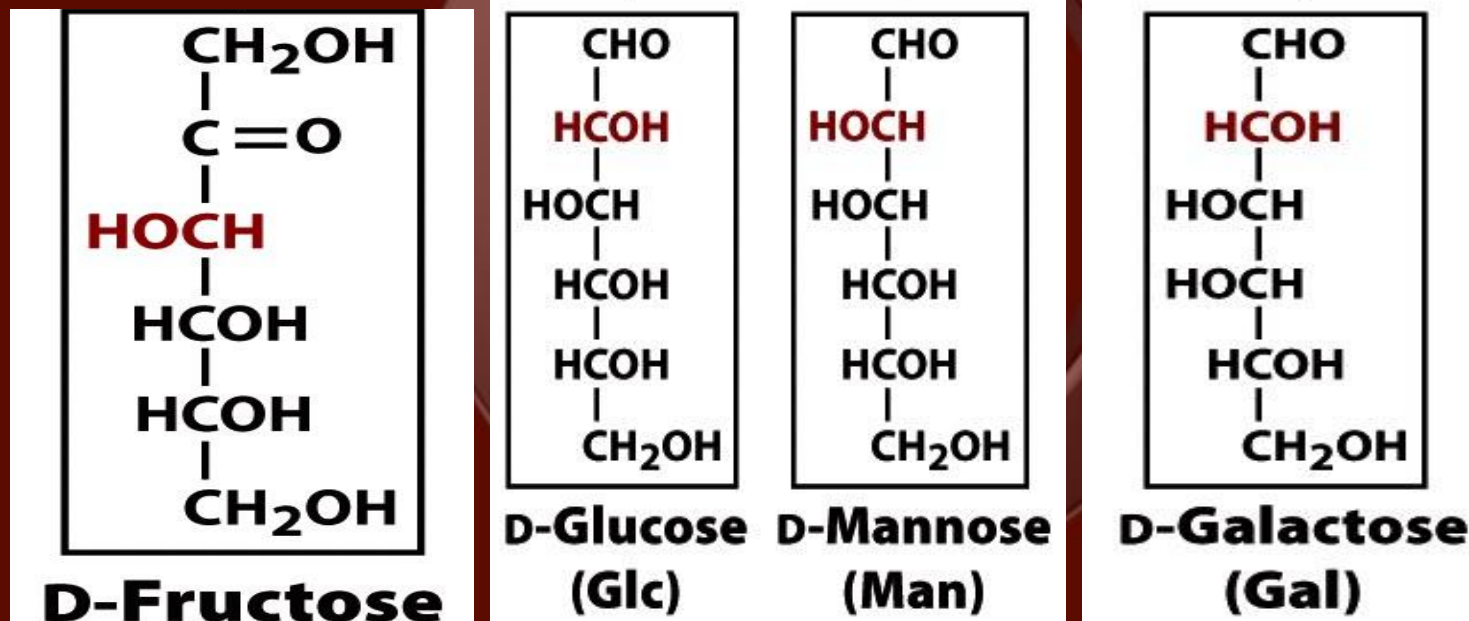
Sugar	Source	Importance	Clinical Significance
D-Glucose	Fruit juices. Hydrolysis of starch, cane sugar, maltose, and lactose.	The "sugar" of the body. The sugar carried by the blood, and the principal one used by the tissues.	Present in the urine (glycosuria) in diabetes mellitus owing to raised blood glucose (hyperglycemia).
D-Fructose	Fruit juices. Honey. Hydrolysis of cane sugar and of inulin (from the Jerusalem artichoke).	Can be changed to glucose in the liver and so used in the body.	Hereditary fructose intolerance leads to fructose accumulation and hypoglycemia.
D-Galactose	Hydrolysis of lactose.	Can be changed to glucose in the liver and metabolized. Synthesized in the mammary gland to make the lactose of milk. A constituent of glycolipids and glycoproteins.	Failure to metabolize leads to galactosemia and cataract.
D-Mannose	Hydrolysis of plant mannans and gums.	A constituent of many glycoproteins.	

Structures of Monosaccharides

- Stereoisomers
- Epimers
- Enantiomers

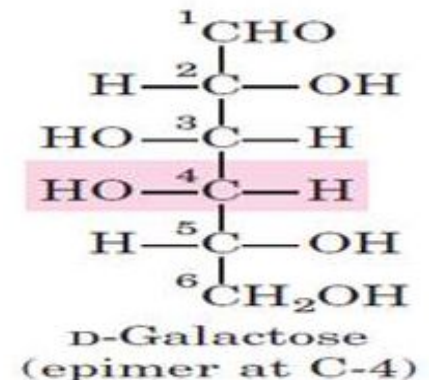
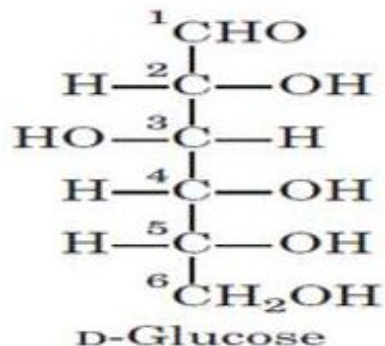
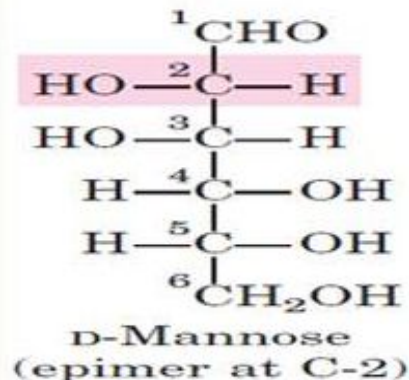
Stereoisomers

- Compounds that have same chemical formula but have different structures or differing in spatial configuration.



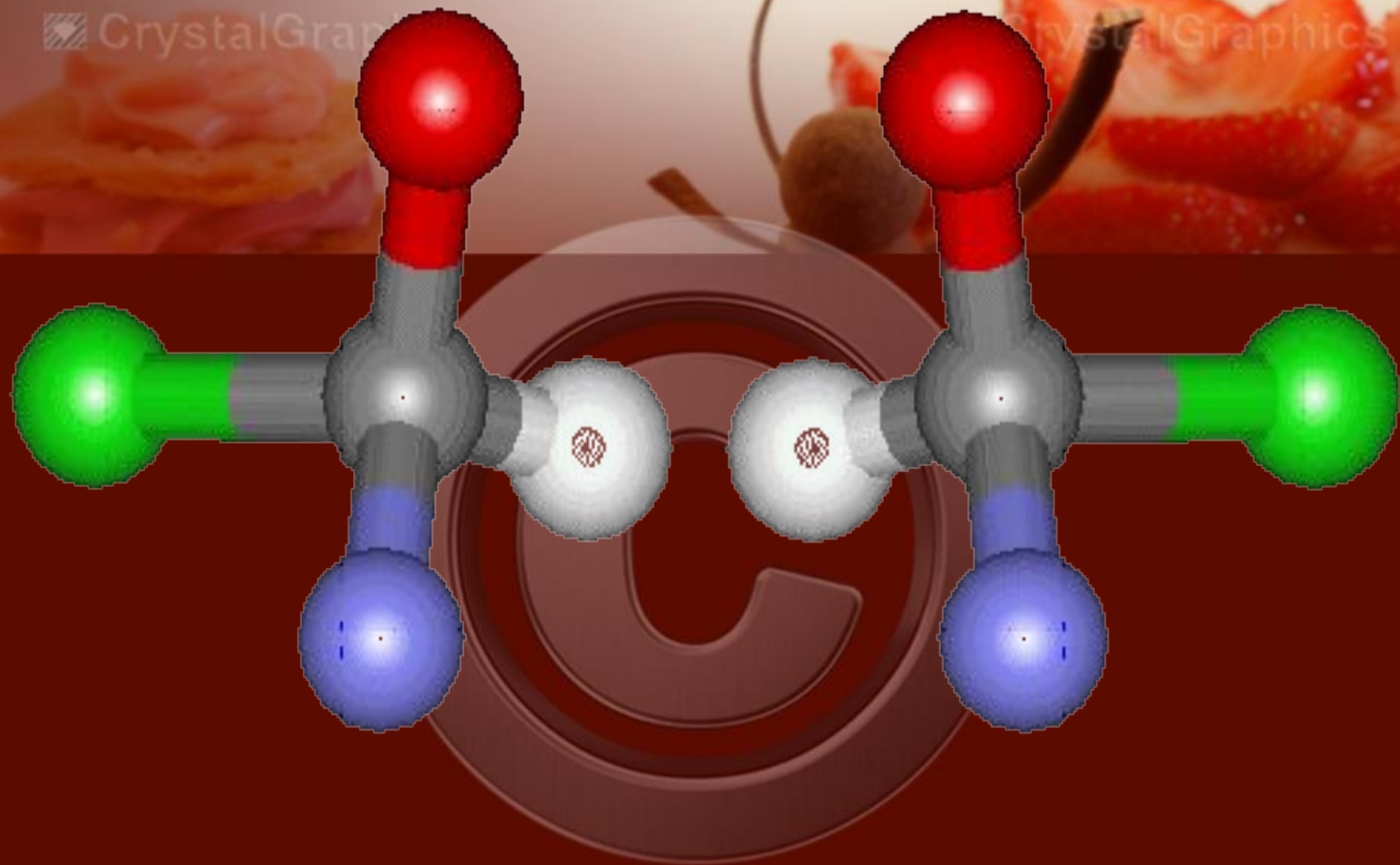
Epimers

- If two monosaccharides differ from each other in their configuration around a single specific carbon atom, they are referred as epimers to each other.
- Glucose & galactose are C4-epimers
- Glucose & mannose are C2-epimers
- Inter-conversion of epimers is known as epimerization, epimerases catalyzes this reaction.



Enantiomers

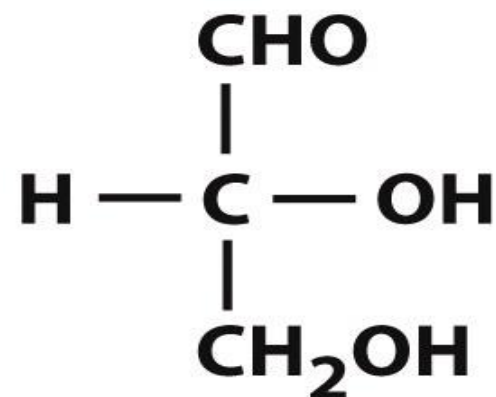
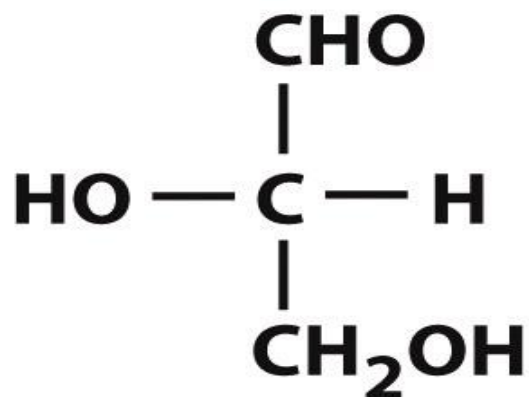
- A special type of isomerism is found in the pairs of structures that are mirror images of each other.
- The two members of the pair are designated as a D- and an L-sugar.
- In the D-isomeric form, the -OH group on the asymmetric carbon is on the right, whereas in the L-isomer, it is on the left.
- Most enzymes are specific for either the D or the L form, but enzymes known as isomerases are able to interconvert D- and L-isomers.



- Asymmetric carbon: Four different groups are attached to the same carbon and farthest from the carbonyl carbon.
- The reference molecule is glyceraldehyde.
- All monosaccharides can be considered as molecules derived from glyceraldehyde by successive addition of carbon atoms.

Enantiomers

Fischer projection



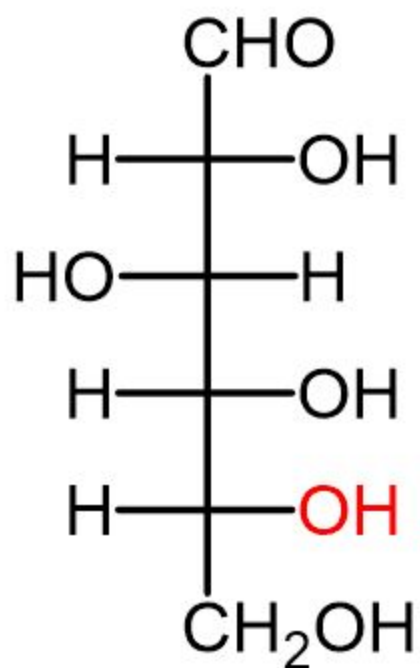
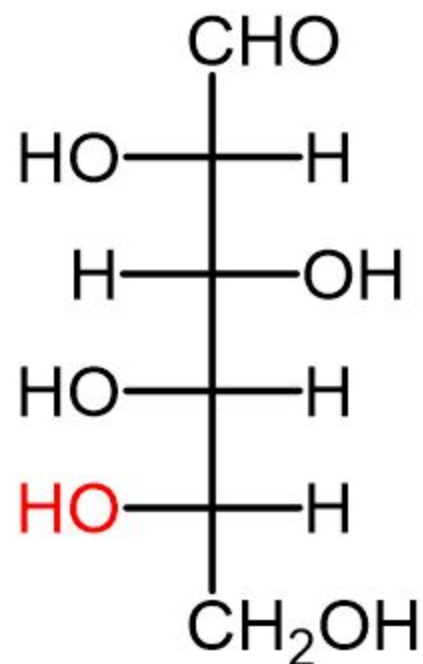
Mirror plane

L-Glyceraldehyde D-Glyceraldehyde

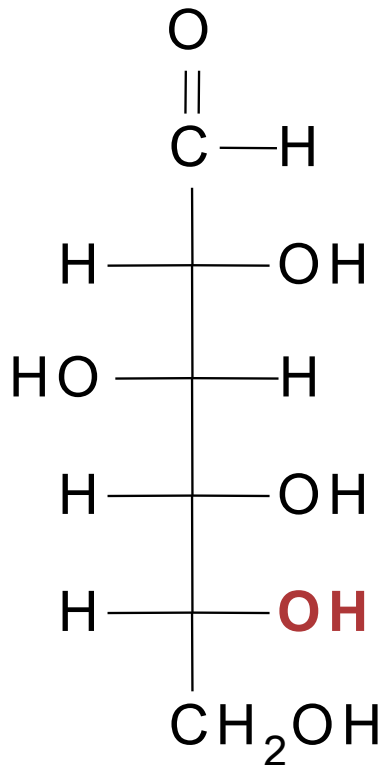
Most sugars are found in the D-stereoisomer.

D and L Isomers of glucose

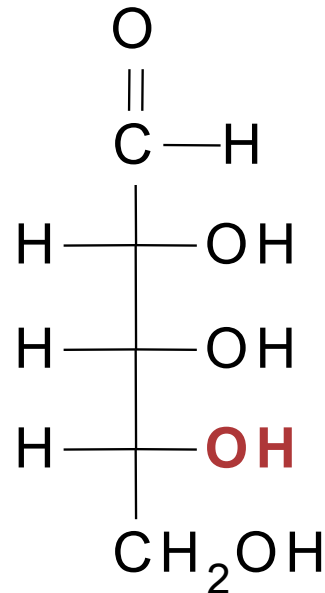
- D and L Isomers are mirror images of each other.
- The spatial orientation of H & OH groups on the C atom (C5 for glucose), adjacent to the terminal primary alcohol carbon determines whether the sugar is D or L Isomer.
- If the OH group is on the right side, the sugar is of D Isomer.
- If the OH group is on the left side, the sugar is of L Isomer.

**D**-glucose**L**-glucose**Enantiomers**

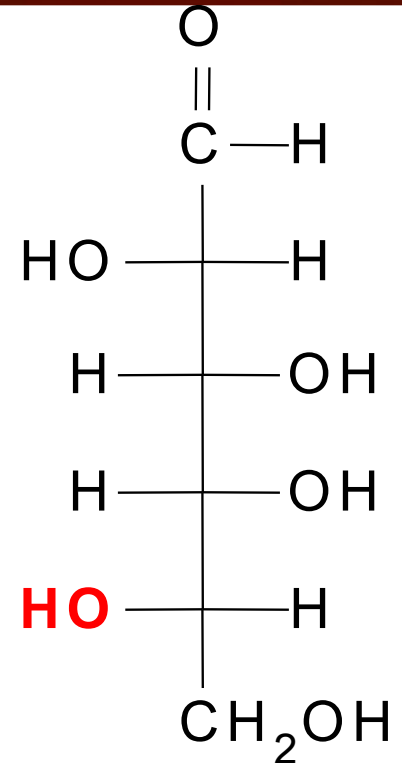
Examples of D and L Isomers of Monosaccharides



D-glucose



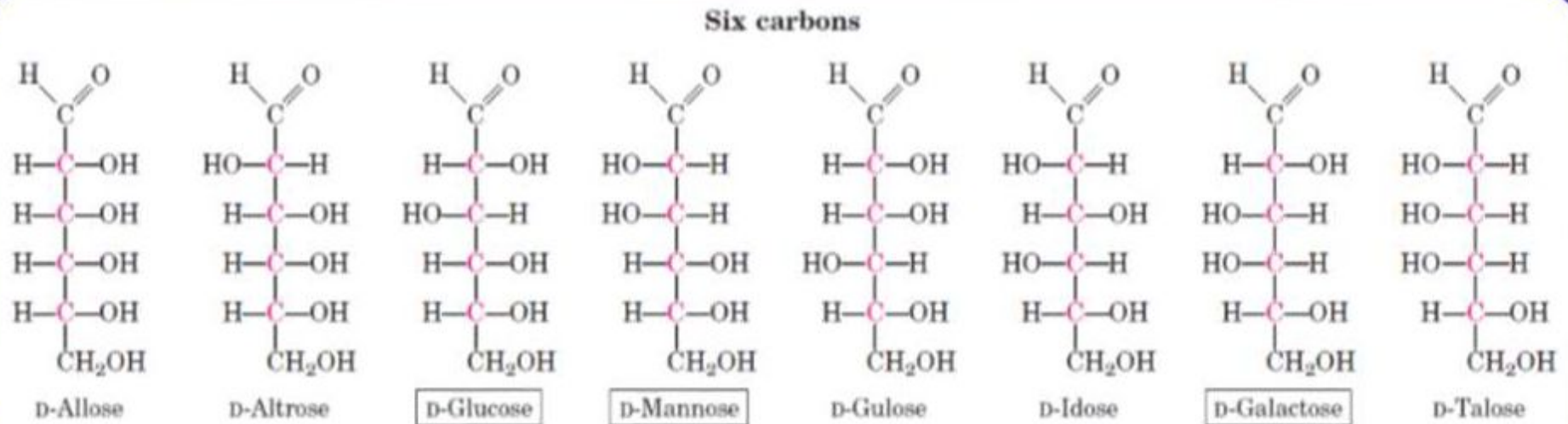
D-ribose



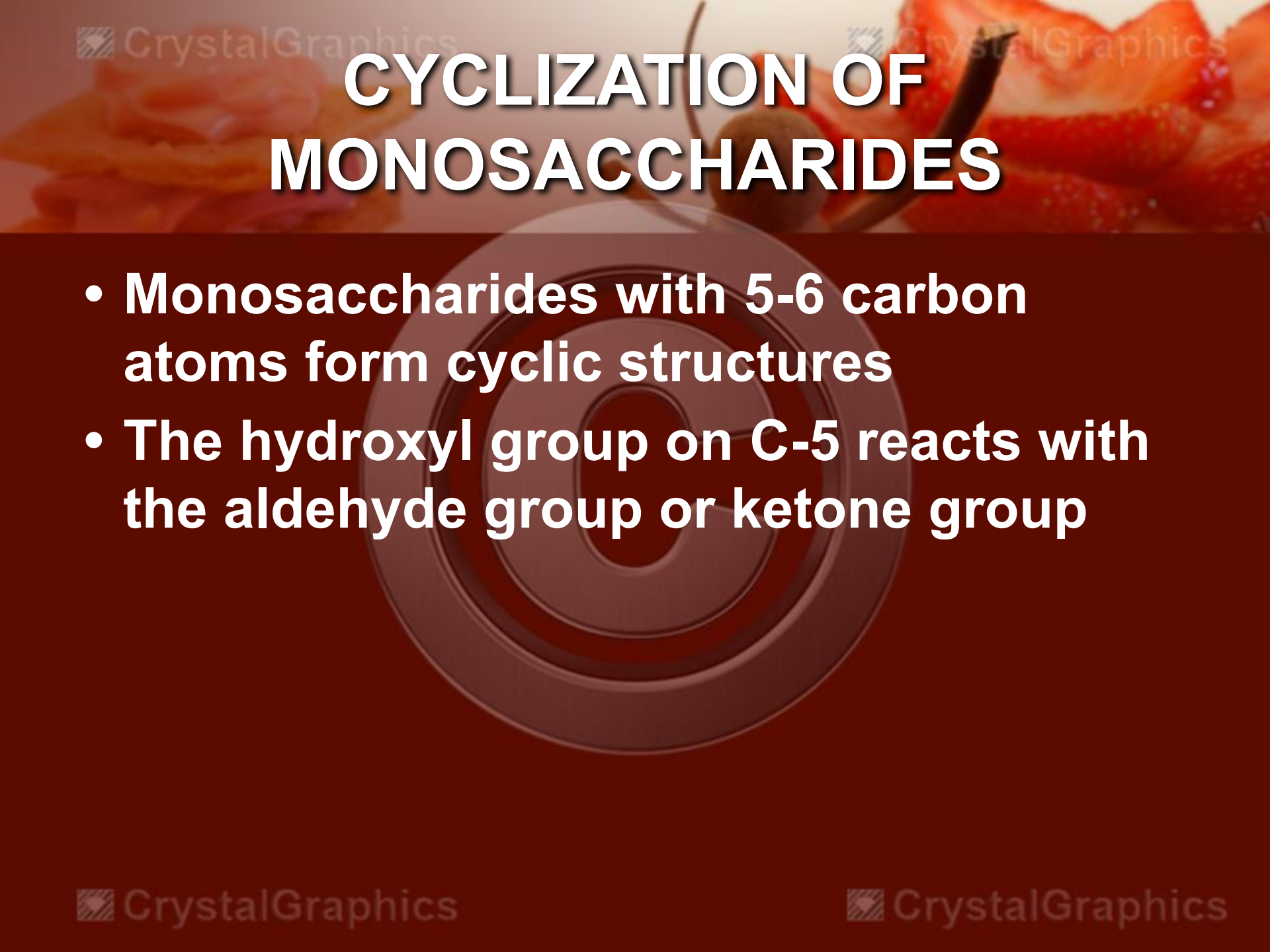
L-galactose

Diastereoisomers

- The term diastereomers is used to represent the stereoisomers that are not mirror images of one another.
- Configurational changes with regard to C2, C3 and C4 will produce eight different monosaccharides.

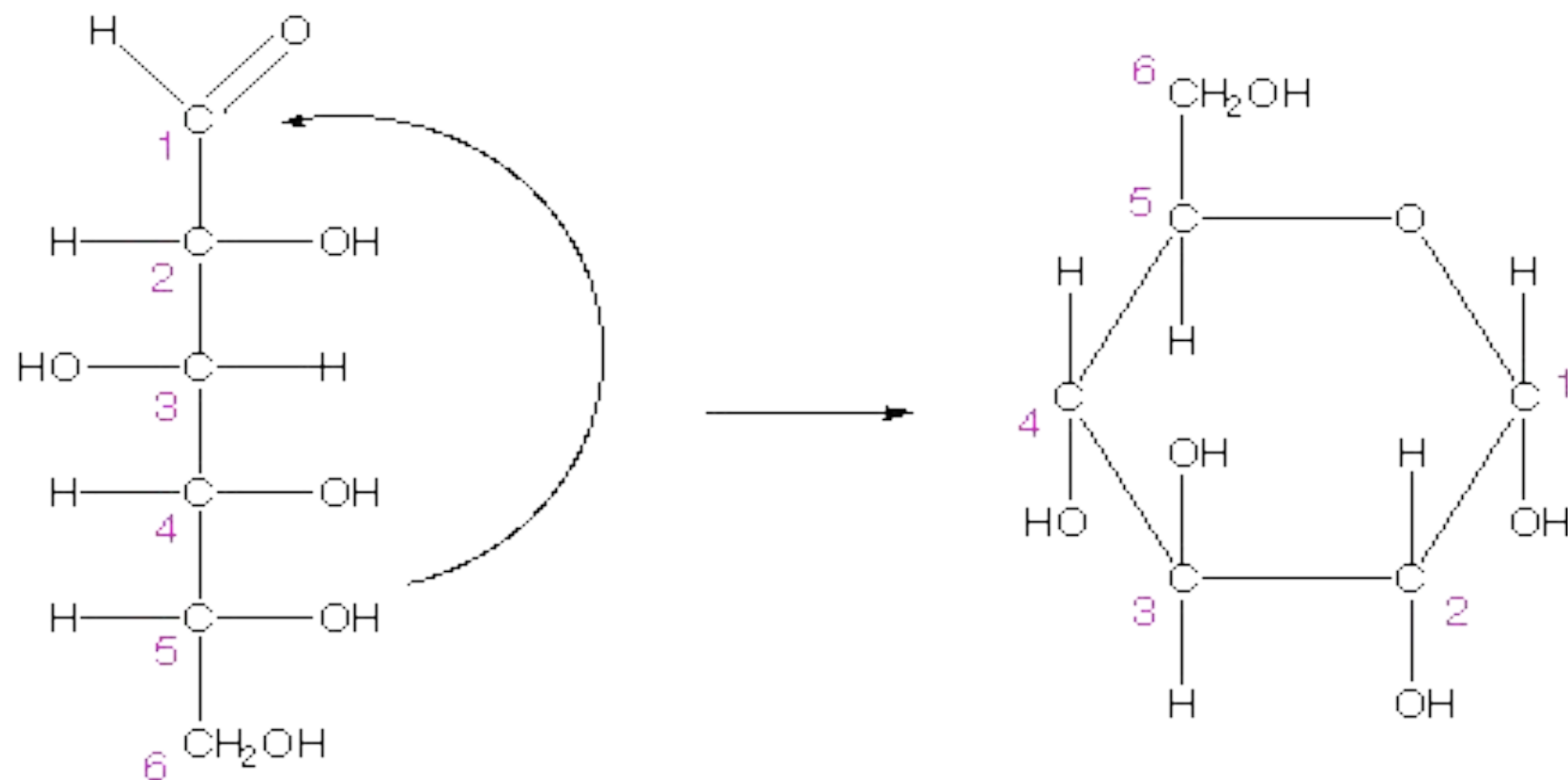


Total D + L forms = 16 isomers of glucose



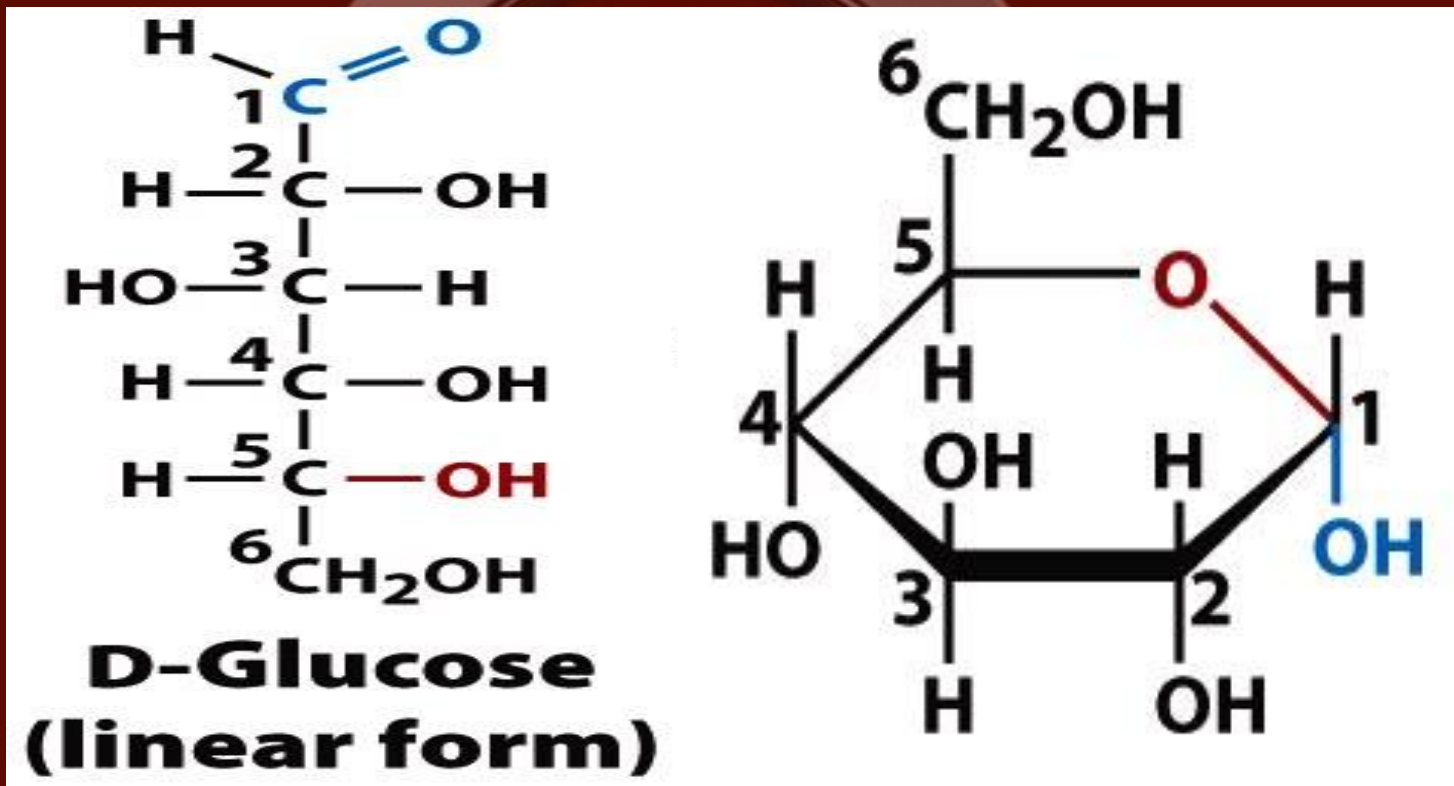
CYCLIZATION OF MONOSACCHARIDES

- **Monosaccharides with 5-6 carbon atoms form cyclic structures**
- **The hydroxyl group on C-5 reacts with the aldehyde group or ketone group**



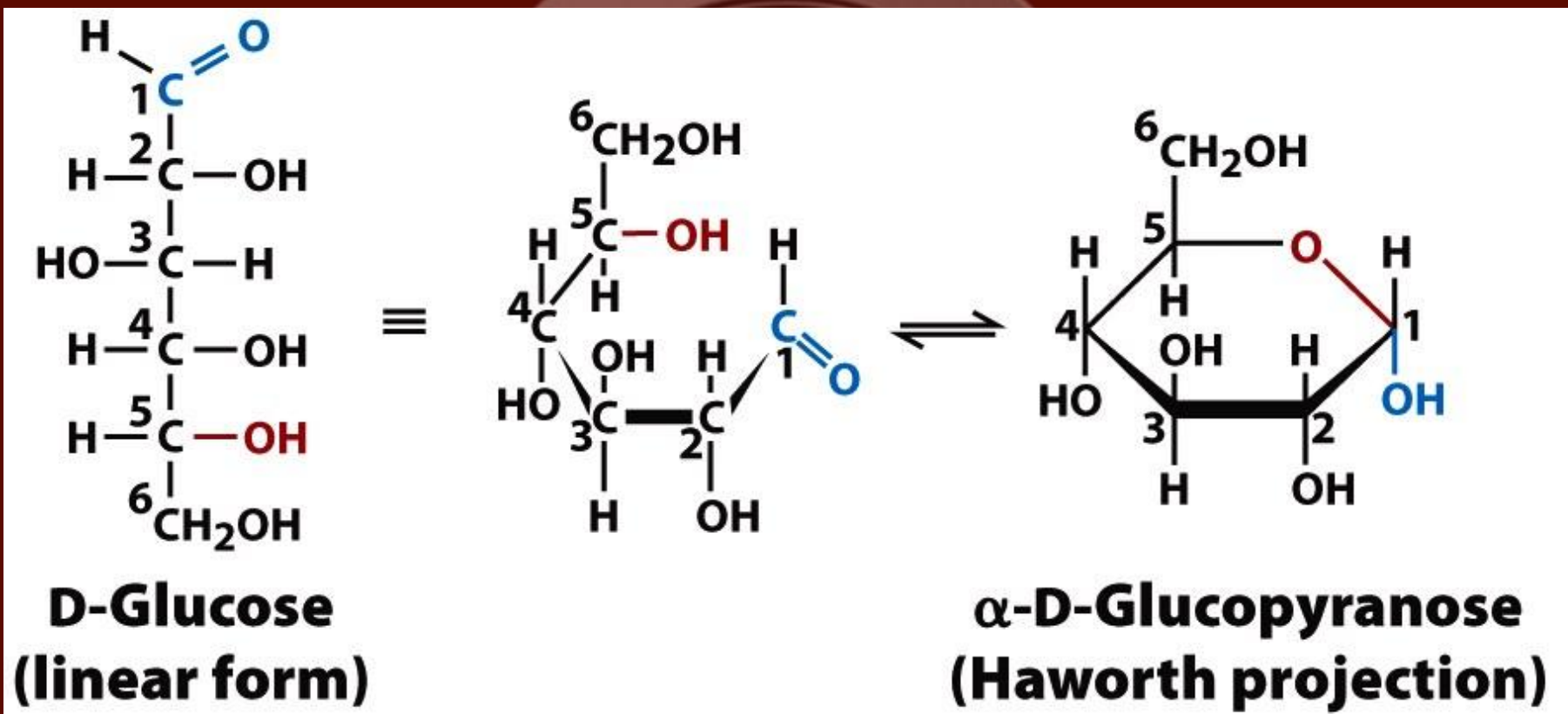
Ways to Draw Monosaccharides

- Fischer Projection - linear
- Haworth Projection - cyclic



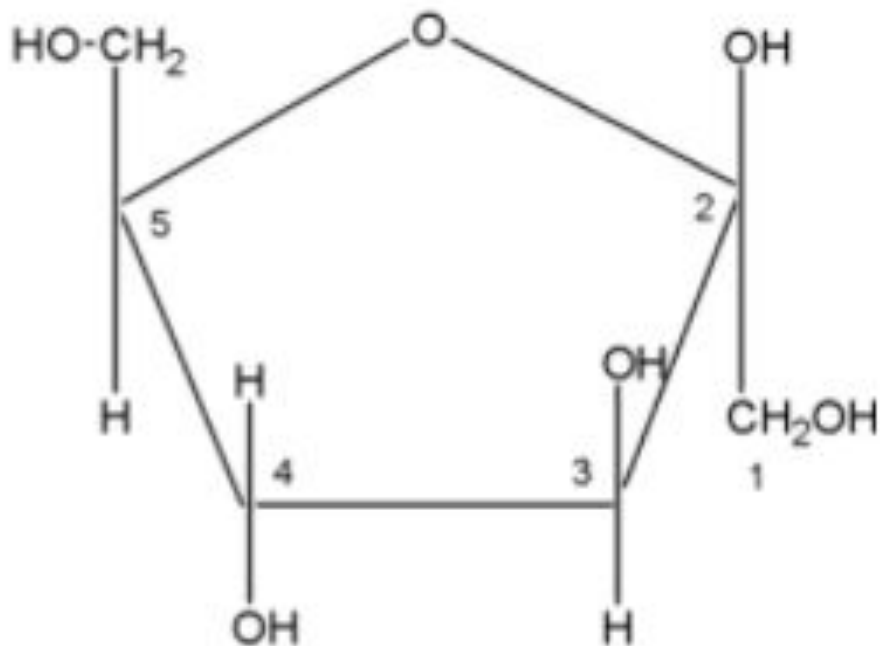
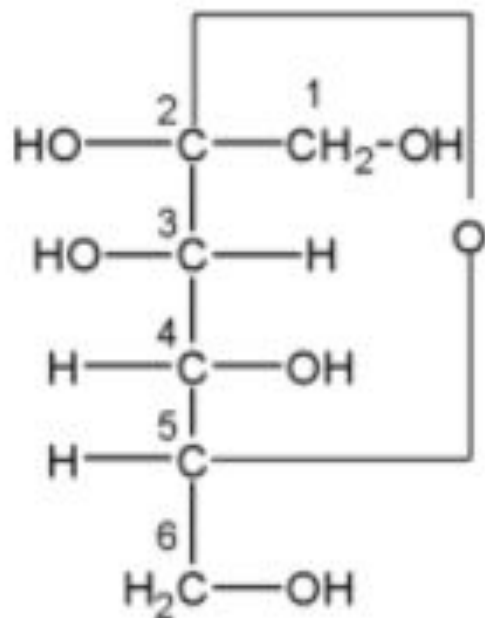
Cyclization of Monosaccharides

Up Down

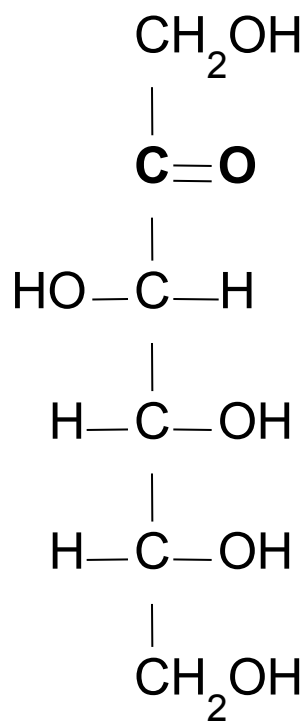


Fructose

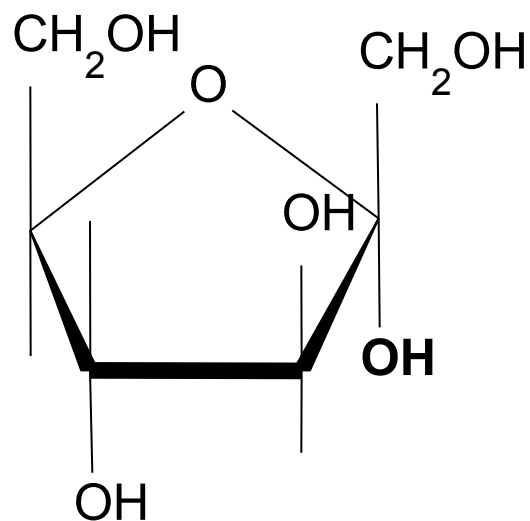
reacts the —OH on C-5 with the C=O on C-2.



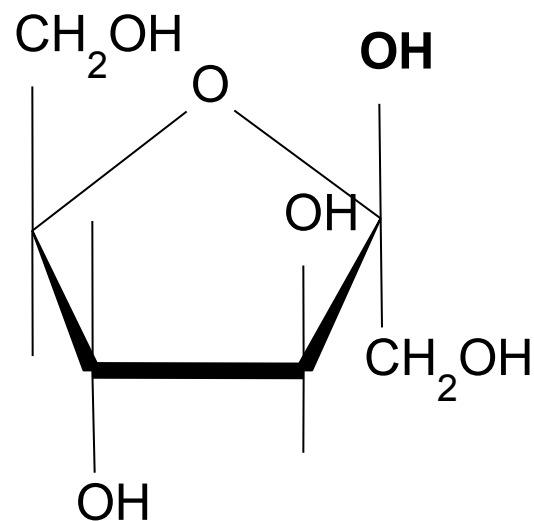
β -D-Fructose



D-fructose



α -D-fructose



β -D-fructose

Anomerism

- Anomers have same composition but differ in the orientation of groups around anomeric carbon atom.
- Anomeric carbon is a carbonyl carbon atom, e.g. 1st carbon atom in glucose is anomeric carbon atom.
- Carbonyl carbon atom becomes asymmetric because of ring structures of monosaccharides in solution thus anomers are encountered in cyclic structures of monosaccharides.

cont

- The alpha & beta cyclic forms of D-glucose are known as anomers. They differ from each other in the configuration only around C1 known as anomeric carbon.
- The hemiacetal (or carbonyl) carbon atom is called the anomeric carbon.

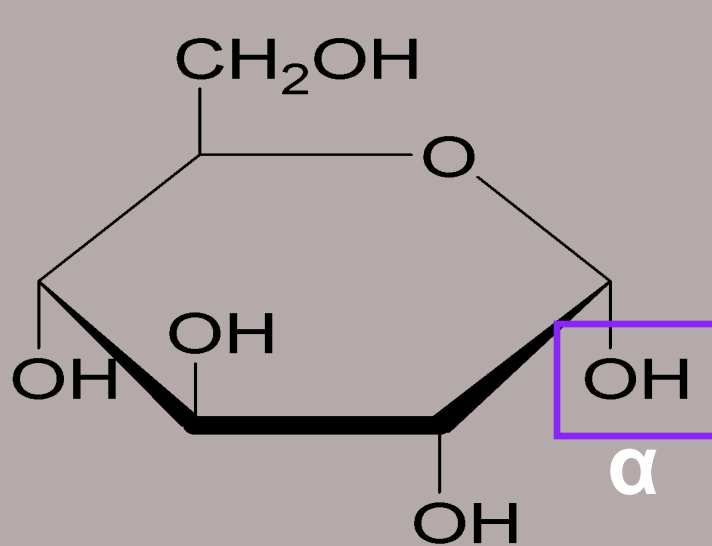
Expression

- Anomers are expressed as α and β forms.
- In α form “OH” group is below the plane (OH group is oriented away from the oxygen atom) .
- In β form “OH” group is above the plane (OH group is oriented towards the oxygen atom)

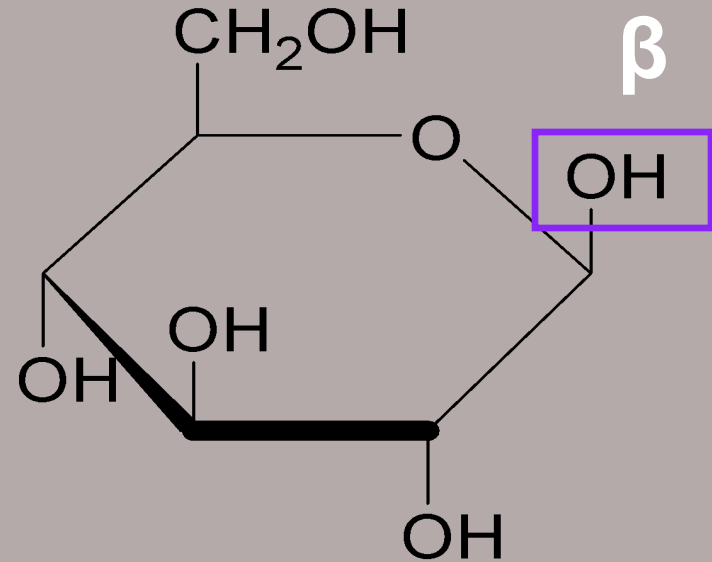
ANOMERIC CARBON

Down OR trans to CH_2OH for the α form.

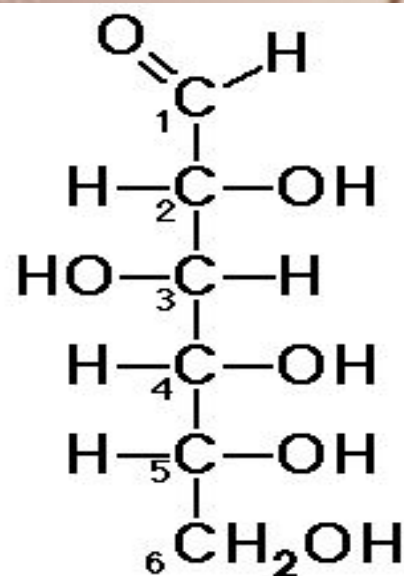
Up OR cis for the β form.



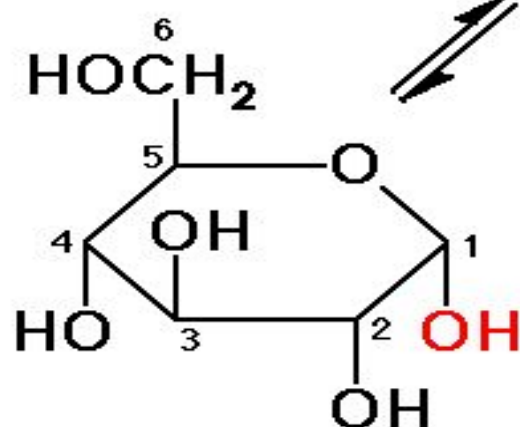
α -D-Glucose



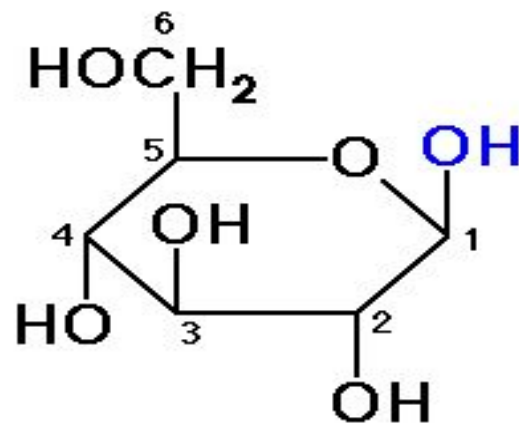
β -D-Glucose



D-glucose



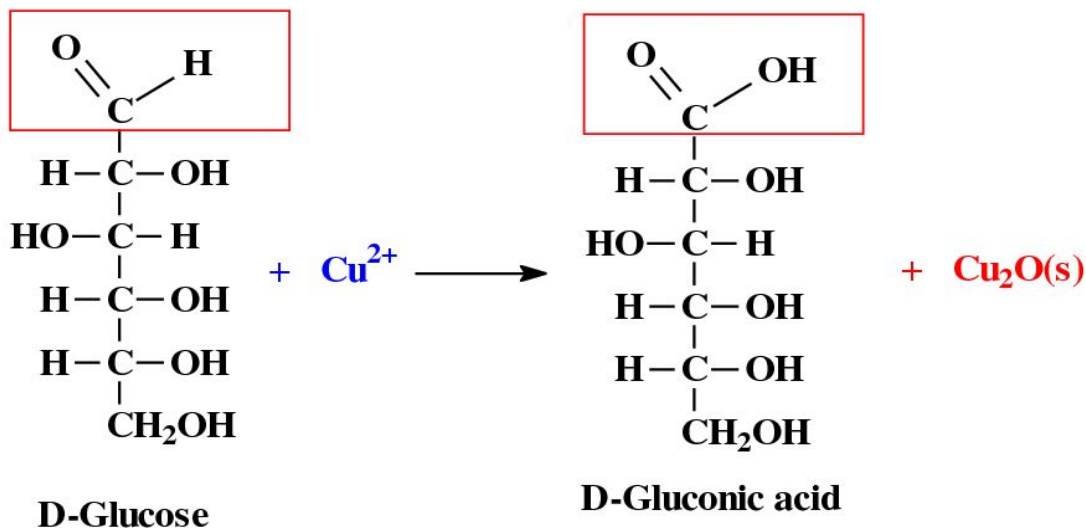
α -D-glucopyranose
(36%)



β -D-glucopyranose
(64%)

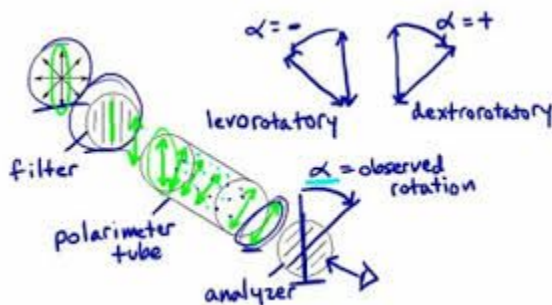
Oxidation of Monosaccharides

- Monosaccharides are reducing sugars if their carbonyl groups oxidize to give carboxylic acids.
- In the Benedict's test, D-glucose is oxidized to D-gluconic acid. Glucose is a reducing sugar.

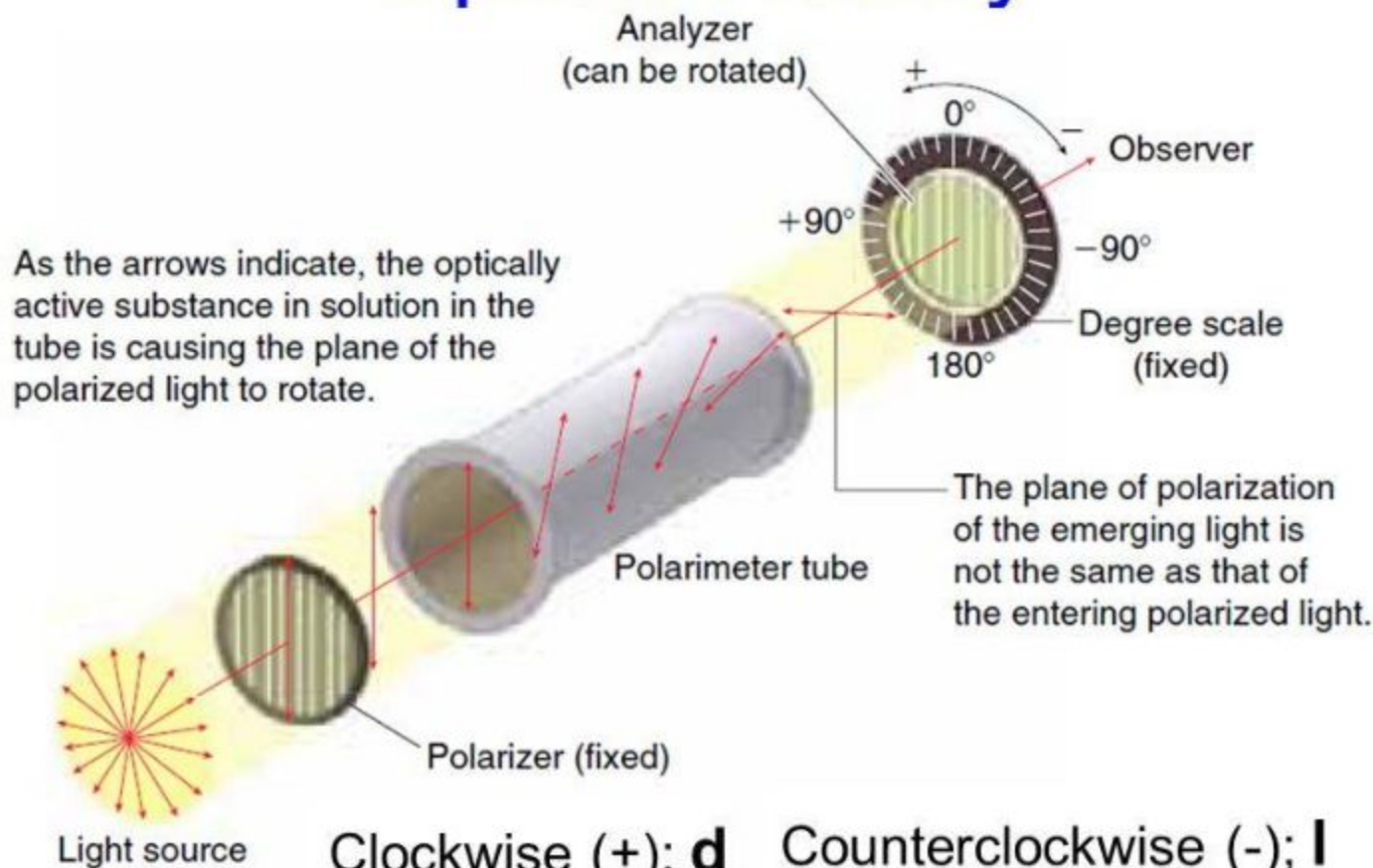


Optical activity

- Optical activity is a characteristic feature of compounds with asymmetric carbon atom.
- When a beam of *polarized light is passes through a solution of an optical isomer, it will be rotated either to the right or left.*
- Depending on the rotation, molecules are called dextrorotatory (+) or levorotatory (-).
- Racemic mixture: If D & L isomers are present in equal concentration, it is known as racemic mixture.
- **NOTE**: Racemic mixture does not exhibit any optical activity.



Optical Activity



Different from absolute (*R,S*) configuration

Glycosides

- When the hemi-acetal group (hydroxyl group of the anomeric carbon) of a monosaccharide is condensed with an alcohol or phenol group, it is called a glycoside.
- The non-carbohydrate group is called aglycone.

Glycoside=sugar group+ nonsugar group
(glycone) (aglycone or genin)

Glycone and aglycone are linked by glycosidic linkage

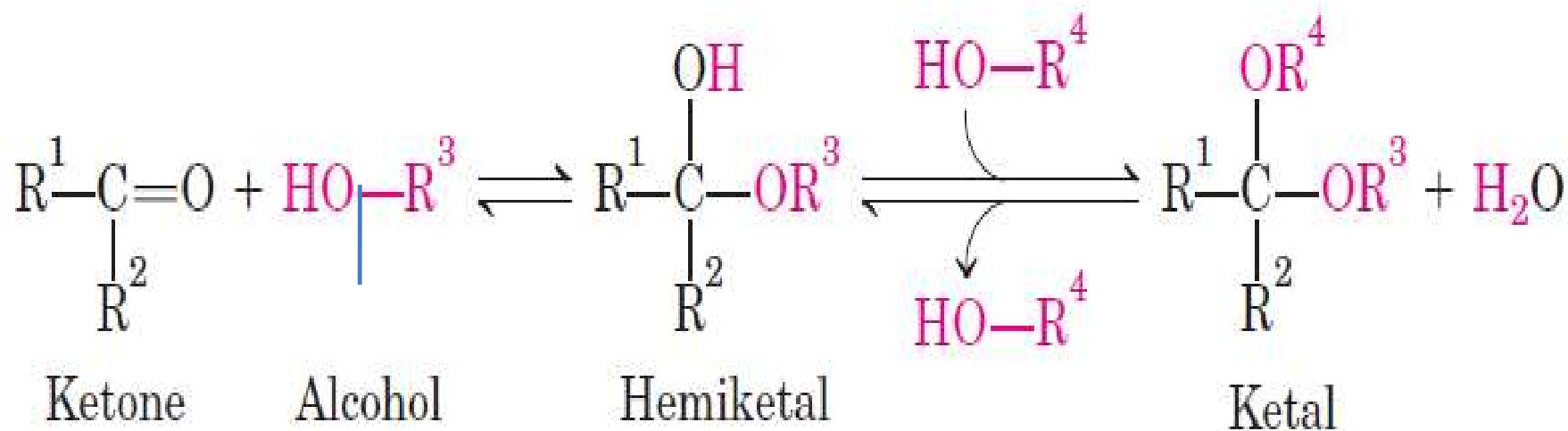
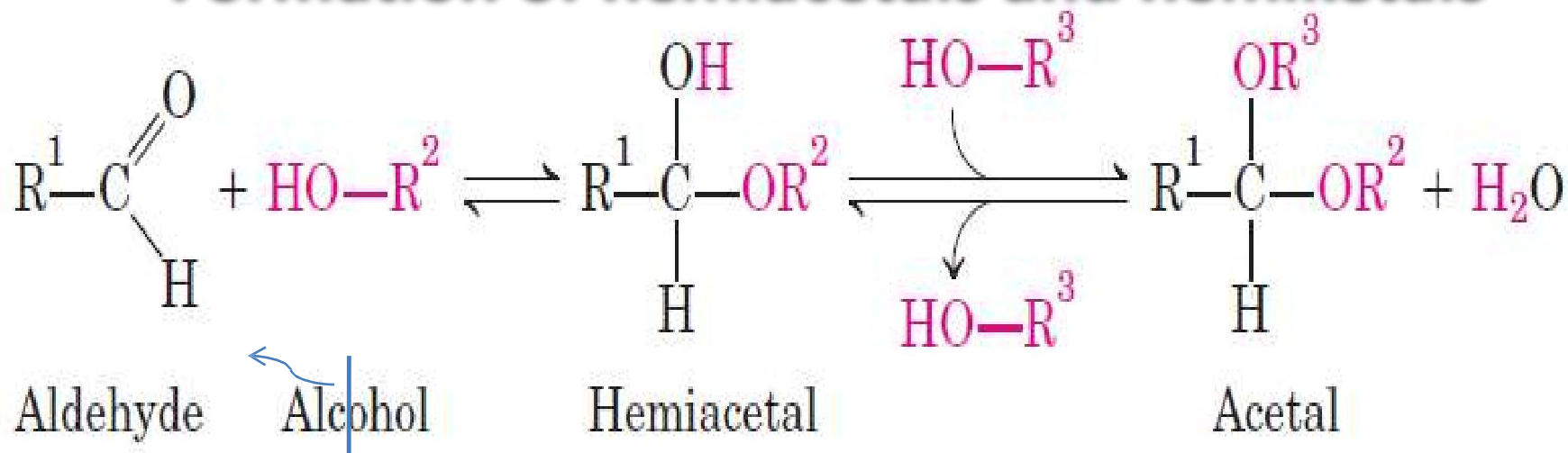
- Atom from the aglycone involved in the glycosidic linkage:

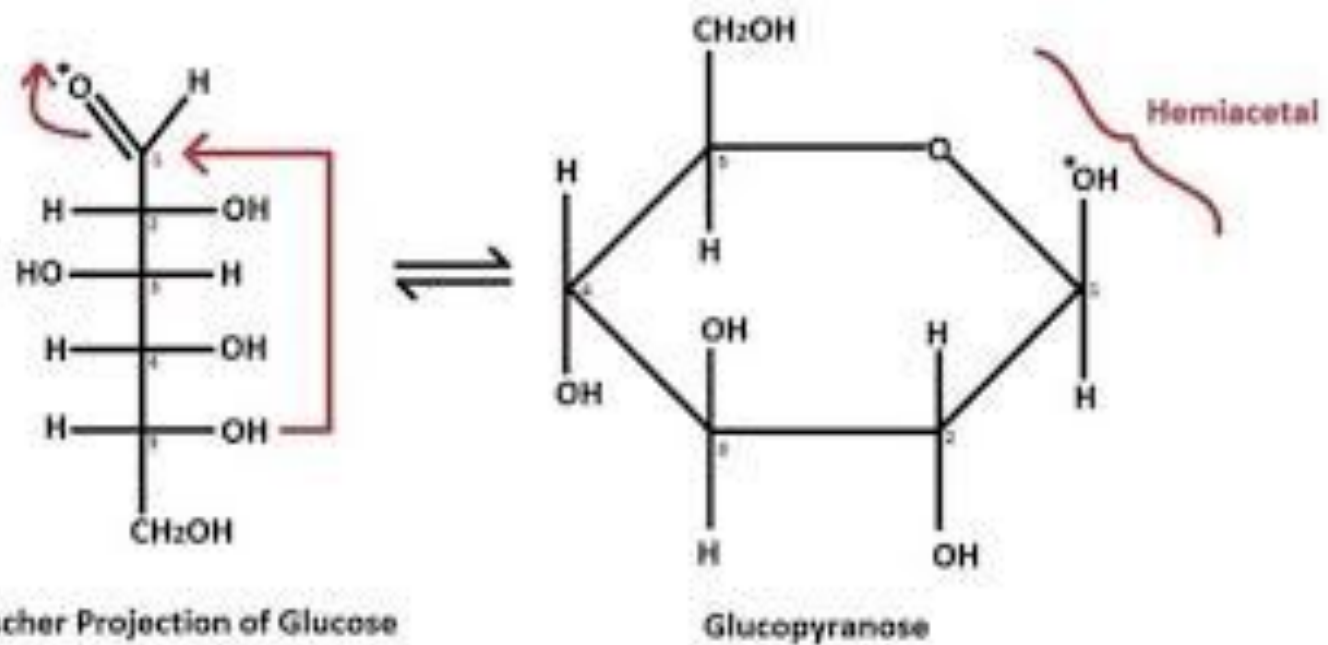
- | | |
|----------------------|--------------|
| • Aglycone- O- Sugar | O-glycosides |
| • Aglycone- C- Sugar | C-glycosides |
| • Aglycone- S- Sugar | S-glycosides |
| • Aglycone- N- Sugar | N-glycosides |

- Number of sugars:

- | | | |
|----------------|-----------|---------------|
| • One sugar | monosides | e.g. Salicin. |
| • Two sugar | Biosides | e.g. Diosmin. |
| • Three sugars | Trisides | e.g. Digoxin. |

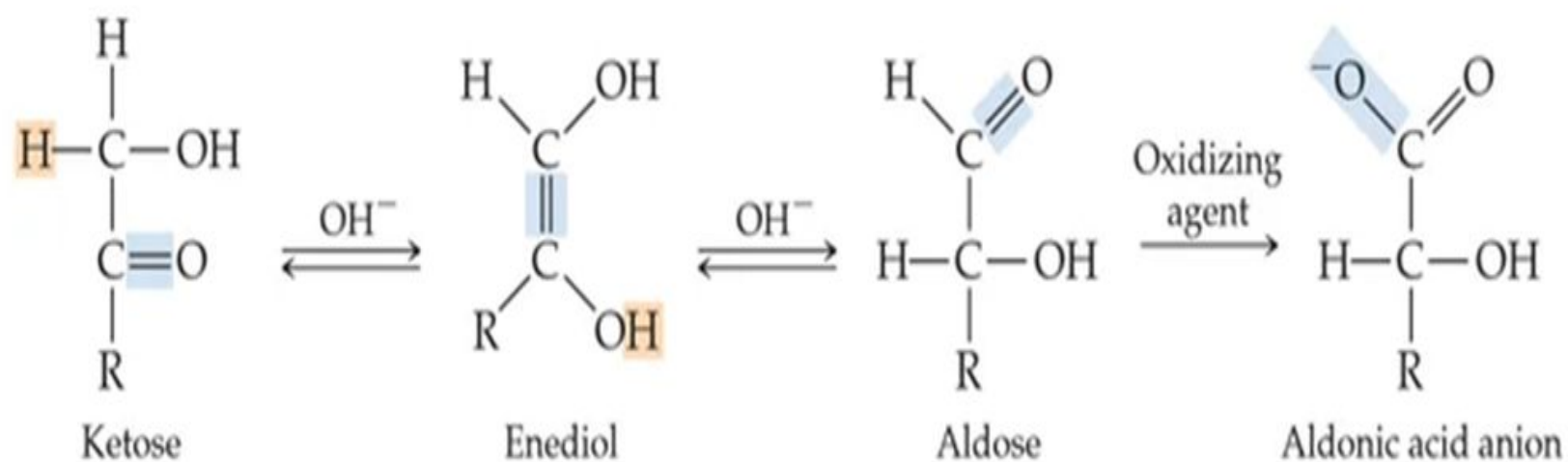
Formation of hemiacetals and hemiketals





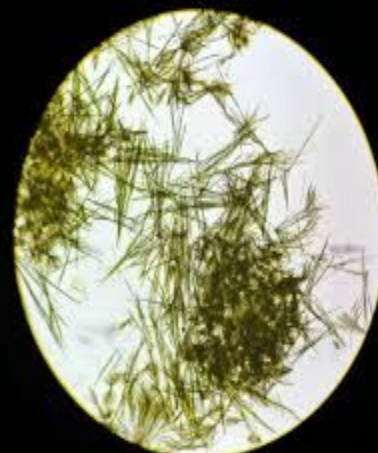
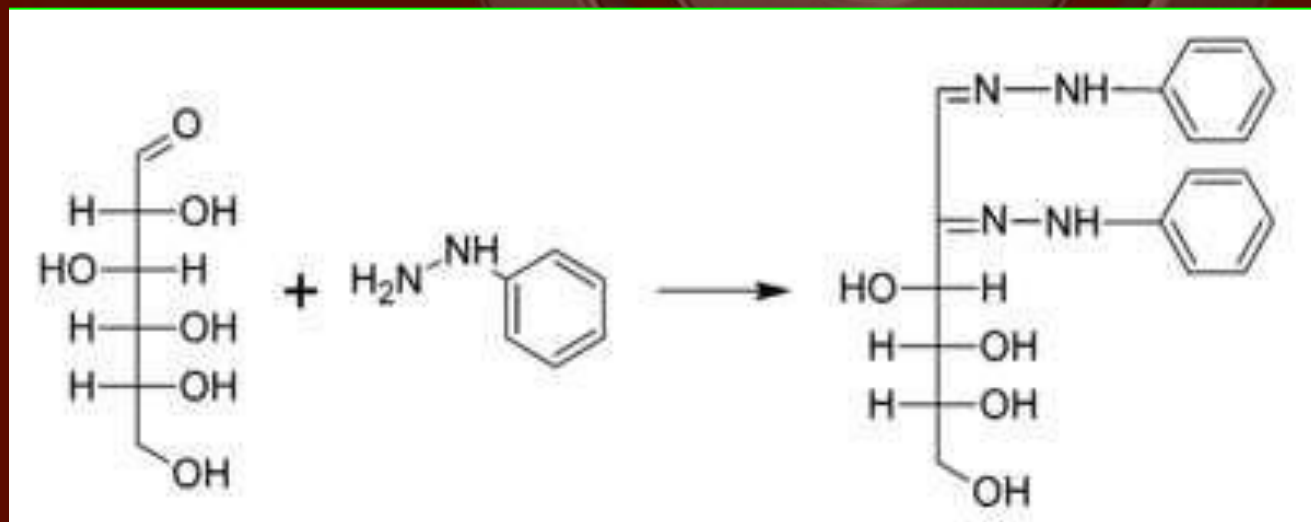
Reactions of Monosaccharides

- **Tautomerization:**
 - The process of shifting a hydrogen atom from one carbon atom to another to produce enediols is known as tautomerization.
 - Crucial in biochemistry for DNA mutation and base-pairing.
- **Reducing properties:**
 - In mild alkaline solutions, carbohydrates containing a free sugar group (aldehyde or ketone) will tautomerise to form enediols.
- **Significance:** Enediols are responsible for the reducing properties of sugars in Benedict's or Fehling's tests, as they readily reduce ions.



Osazone formation

- All reducing sugars will form osazones with excess of phenylhydrazine when kept at boiling temperature.
- Osazones are insoluble.
- Osazones of individual sugars have characteristic crystal form
- The differences in glucose, fructose and mannose are dependent on 1st and 2nd C & this difference is masked when Phenylhydrazine reacts with these two carbons.
- So, Glucose, Fructose and Mannose give broom shaped osazones.



Needle shaped/ hedgehog shaped osazones formed by glucose

Oxidation of Sugars

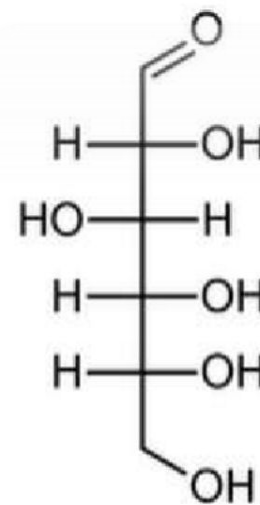
In mild oxidation, the aldehyde group is oxidized to carboxyl group to produce ALDONIC acids.

Glucose $\rightarrow$ Gluconic acid Mannose $\rightarrow$

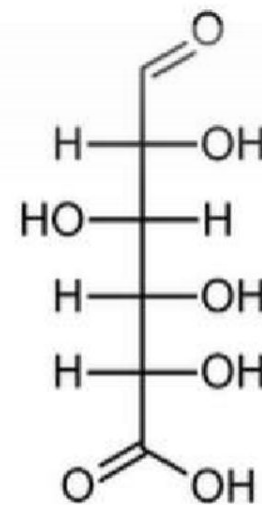
Mannonic acid Galactose $\rightarrow$ Galactonic acid

When Aldehyde group is protected, and the molecule is oxidized at the last C to COOH producing URONIC acids.

Imp- Glucuronic acid is used in body to synthesize heteropolysaccharides and also for conjugation of various substances.



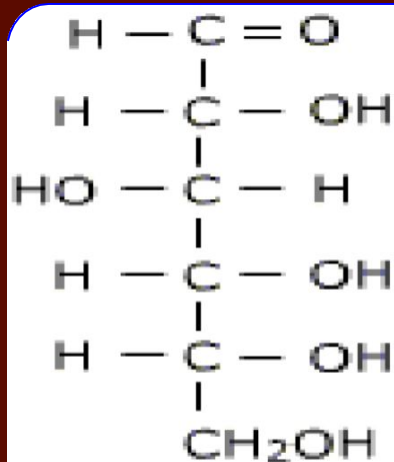
Glucose



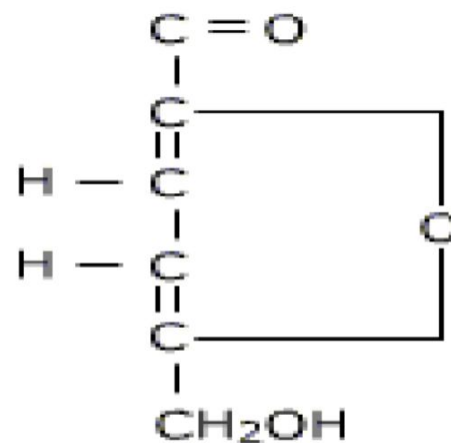
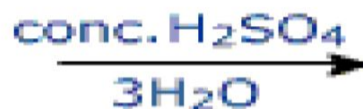
Glucuronic acid

Furfural formation

- Monosaccharides when treated with concentrated H_2SO_4 undergoes dehydration with the removal of 3 molecules of water.
 - Hexoses give hydroxymethyl furfural and pentoses give furfural.
 - Furfurals condense with phenolic compounds to give various colors.
- E.g. Molisch's test: General test for carbohydrates (H_2SO_4 and α -naphthol)
- Rapid Furfural and Seliwanoff's test: Tests for presence of keto group



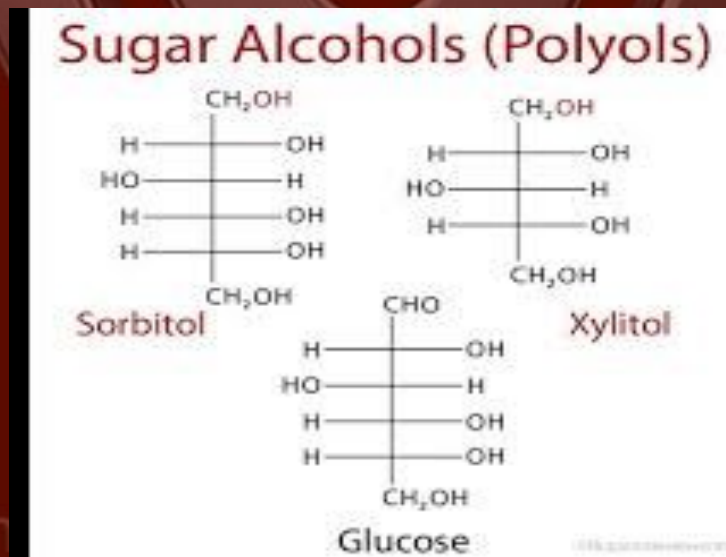
D - Glucose



Hydroxy methyl furfural

Reduction to form alcohols

- The reduction of sugars to alcohols (specifically sugar alcohols) involves the conversion of the carbonyl group (aldehyde or ketone) of a sugar into a primary or secondary hydroxyl group (-OH).
- D-Glucose $\square$ D-Sorbitol $\square$ D-Fructose $\square$ D-Mannitol
- Sorbitol and Mannitol are used to identify bacterial colonies.
- Mannitol is used to reduce intracranial pressure by forced diuresis.



Formation of esters

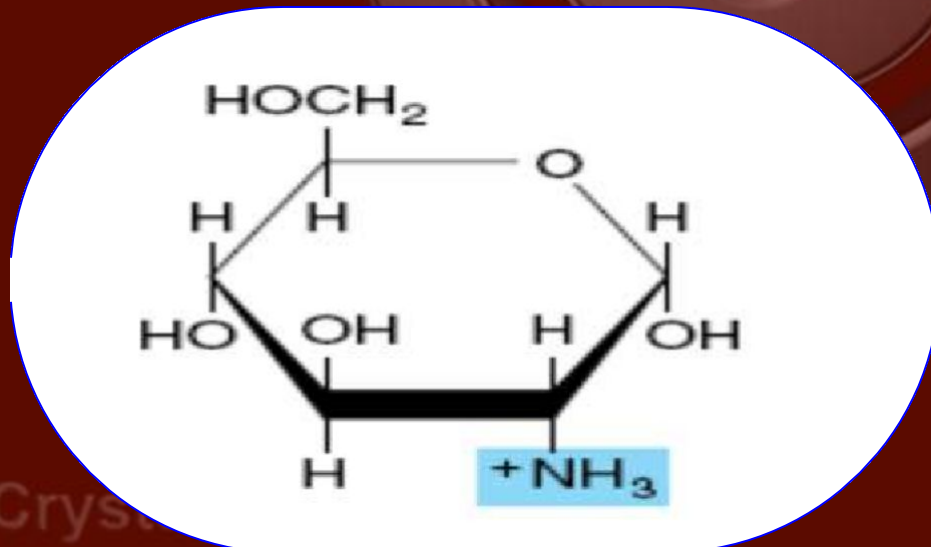
- Hydroxyl groups of sugars can be esterified to form acetates, propionates, benzoates, etc
- Sugar phosphates are of great biological importance.
- Metabolism of sugars inside the body starts with phosphorylation.
e.g Glucose 6- PO_4

Amino sugars

- Amino groups may be substituted for hydroxyl groups of sugars to give rise to amino sugar.
- Generally the group is added to the second C of hexoses.
- They are non-reducing and do not form osazones
- They are found in GAGS, glycoproteins, proteoglycans
- Abbreviations:-

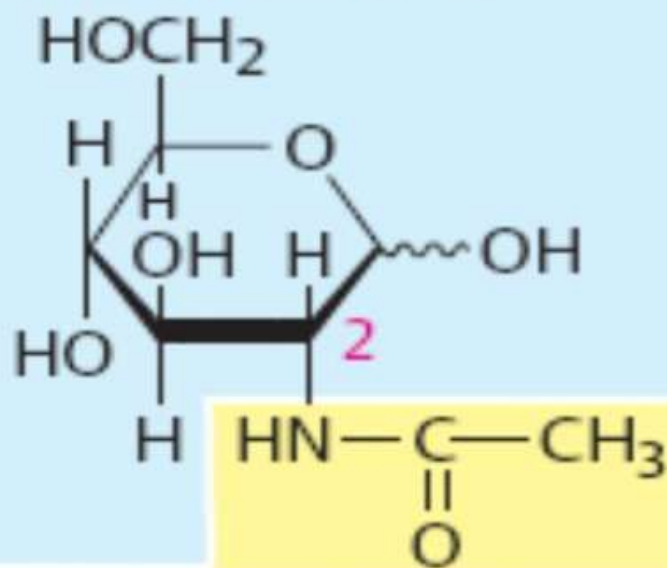
GluNac = N-acetyl-glucosamine

GalNac = N-acetyl-galactosamine

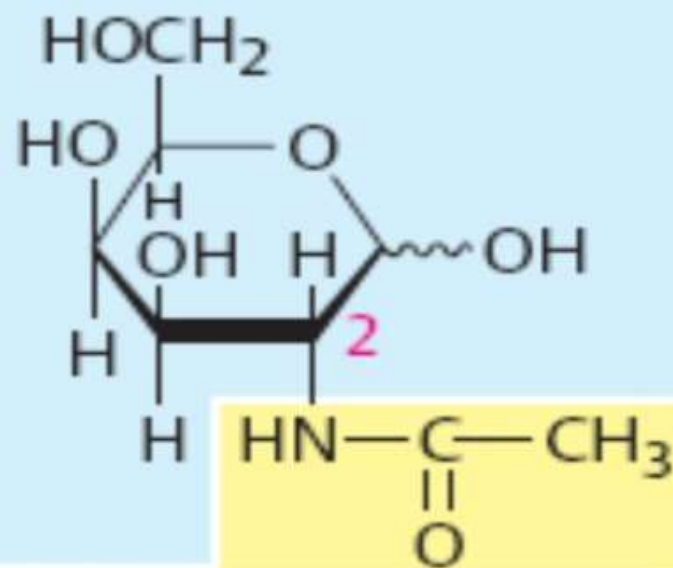


GLUCOSAMINE or 2 amino-D-glucopyranose (α form)

N-Acetyl-D-glucos-
amine (GlcNAc)



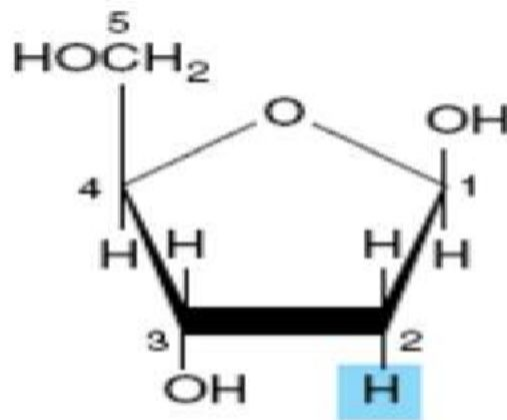
N-Acetyl-D-galac-
tosamine (GalNAc)



- The amino group may be further acetylated to produce N-acetylated sugars like N-acetyl glucosamine (GlcNAc) or N-acetyl galactosamine (GalNAc) which are important constituents of glycoproteins and MPS

Deoxy Sugars

- Oxygen of the hydroxyl group may be removed to form deoxy sugars.
- They are non reducing.
- Don't form osazones.
- Deoxyribose is present in DNA



2-Deoxy-D-ribofuranose (β form).

Thank you